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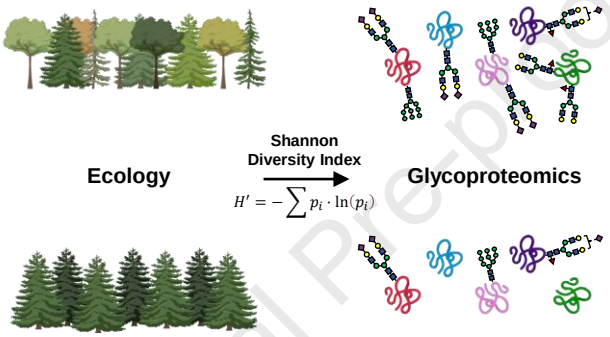
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Running Title: Protein glycosylation heterogeneity in renal tumors

ABBREVIATIONS

A2M = α-2-macroglobulin

ACAN = aggrecan core protein

AHSG = α-2-HS-glycoprotein

AGC = automatic gain control

AOC3 = amine oxidase [copper-containing] 3

ALG1 = chitobiosyldiphosphodolichol β-mannosyltransferase

APMAP = adipocyte plasma membrane-associated protein

APOD = apolipoprotein D

AZGP1 = zinc-α-2-glycoprotein

B3GNT8 = N-acetyllactosaminide β-1,3-N-acetylglucosaminyltransferase 8

BCA = bicinchoninic acid

BSA = bovine serum albumin

cDNA =complementary DNA

CLPTM1L = lipid scramblase CLPTM1L

CLU = clusterin

COL12A1 = collagen α-1(XII) chain

COL6A1 = collagen α-1(VI) chain

DDA = data-dependent acquisition

DPAGT1 = UDP-N-acetylglucosamine--dolichyl-phosphate N-acetylglucosaminephosphotransferase

ECM = extracellular matrix

ER = endoplasmic reticulum

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F2 = prothrombin

FDR = false discovery rate

FN1 = fibronectin

FUT8 = α-(1,6)-fucosyltransferase

FUT11 = GDP-fucose protein O-fucosyltransferase 4

FT = Fourier transform

Fuc = fucose

FUT6 = 4-galactosyl-N-acetylglucosaminide 3-α-L-fucosyltransferase

GANAB = glucosidase II α subunit, neutral α-glucosidase AB

GEO = Gene Expression Omnibus

GFUS = GDP-L-fucose synthase

GO = gene ontology

GSMs = glycopeptide spectrum matches

HCD = higher-energy collision-induced dissociation

Hex = hexose

HexNAc = N-acetylhexosamine

HSP90B1 = endoplasmin

HSPG2 = basement membrane-specific heparan sulfate proteoglycan core protein

HYOU1 = hypoxia up-regulated protein 1

IGHA2 = immunoglobulin heavy constant α 2

IRB = institutional review board

ITGA1 = integrin α-1

ITGA3 = integrin α-3

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ITGAV = integrin α-V

LAMA5 = laminin subunit α-5

LAMP = lysosomal-associated membrane protein

LC-MS/MS = liquid chromatography-tandem mass spectrometry

MAN2B1 = lysosomal α-mannosidase

MGAT1 = α-1,3-mannosyl-glycoprotein 2-β-N-acetylglucosaminyltransferase

MMRN2 = multimerin-2

MS = mass spectrometry

N(number) = asparagine(residue number)

NeuAc = sialic acid

OCT = optimal cutting temperature

ORM1 = α-1-acid glycoprotein 1

PCR = polymerase chain reaction

PE = paired-end

PLXNB2 = plexin-B2

POFUT4 = protein O0fucosyltransferase 4

POSTN = periostin

PSAP = prosaposin

RCC = renal cell carcinoma

RNA = ribonucleic acid

RNA-seq = RNA-sequencing

SAX-ERLIC = strong anion exchange-electrostatic repulsion hydrophilic interaction chromatography

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SERPINA3 = α-1-antichymotrypsin

SERPING1 = plasma protease C1 inhibitor

Siglec = sialic acid-binding immunoglobulin-type lectin

SPE = solid phase extraction

ST3GAL1 = CMP-N-acetylneuraminate-β-galactosamide-α-2,3-sialyltransferase 1

ST3GAL2 = CMP-N-acetylneuraminate-β-galactosamide-α-2,3-sialyltransferase 2

ST3GAL3 = CMP-N-acetylneuraminate-β-1,4-galactoside α-2,3-sialyltransferase

ST3GAL5 = lactosylceramide α-2,3-sialyltransferase

TCEP = tris(2-carboxyethyl)phosphine

TFA = trifluoroacetic acid

THBS1 = thrombospondin-1

TMED9 = transmembrane emp24 domain-containing protein 9

TPM = transcripts per million

UHPLC = ultra-high performance liquid chromatography

uniLASSO = univariate least absolute shrinkage and selection operator

UGT1A9 = UDP-glucuronosyltransferase 1A9

VCAN = versican core protein

KEYWORDS

Protein glycosylation, renal, tumors, Shannon diversity index, mass spectrometry, glycoproteomics

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ABSTRACT

Extensive changes in glycosylation are observed in many malignancies, but limited information is available about the modified proteins and sites. We used intact glycoproteomics to map N-glycosites in 32 surgically resected renal tumors and paired adjacent normal tissues. We identified 853,340 total and 54,354 unique N-linked glycopeptides from 1,940 glycoproteins containing 220 glycan compositions. The Shannon diversity index was used to characterize glycosylation diversity and identified tumor-associated differences enriched in glycoproteins mapping to cell-adhesion and immunity processes. Although more glycopeptides were identified in tumors, whole-glycoproteome Shannon diversity did not differ significantly between tissue types; tumor-associated differences were concentrated at specific glycoproteins and glycosites. . In tumors, complex fucosylated and complex sialylated glycans were more common. RNA-seq and uniLASSO-based integration nominated glycosylation-related transcripts associated with site-specific glycan-category changes. Changes in total protein abundance were moderately associated with changes in Shannon diversity at the protein ($R^2 = 0.290$) and glycosite ($R^2 = 0.279$) levels, but most of the linear variation was not accounted for by protein abundance alone. Diversity changes were also concentrated in functional and structural protein domains. The results provide prioritized candidates and hypotheses for independent validation and mechanistic studies of protein glycosylation changes in renal tumors.

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INTRODUCTION

Protein glycosylation, the covalent attachment of carbohydrate, or glycan, structures onto specific sites of the protein backbone, is the most abundant and diverse post-translational modification, occurring in >50% of human proteins (1). It employs hundreds of enzymes, including glycosyltransferases and glycosidases, and 16 distinct pathways that are responsible for the assembly, processing, and turnover of glycans (2, 3). Since protein glycosylation plays pivotal roles in protein folding, localization, stability, activity, and functional interactions, aberrant glycosylation, such as increased sialylation and fucosylation, is implicated in a variety of diseases, including cancer (4, 5). Altered glycosylation patterns, at least in part due to dysregulated expression of glycosyltransferases and glycosidases, are increasingly recognized as fundamental drivers of not only tumorigenesis but also tumor migration, metastasis, immune evasion, and drug resistance (6-10). As such, these aberrant glycosylation signatures offer insights into cancer biology and serve as valuable diagnostic markers and potential therapeutic targets across a range of malignancies, particularly in cancer (11-13). Advances in targeting cancer-associated glycosylation changes using novel strategies hold great promise in improving the specificity and efficacy of cancer therapeutics (14-17).

Aberrant glycosylation has been observed in several kidney diseases, including renal cell carcinoma (RCC), one of the most common malignancies of the urinary system that accounts for 2-3% cancer deaths (18, 19). Similar to cancers in the breast, lung, and colon, protein glycosylation plays critical roles in RCC tumorigenesis and progression through dysregulating signaling pathways involved in the metabolism, growth, survival, migration, and metastasis of cancer cells, as well as their ability to induce angiogenesis and immune evasion (20-23). Much of this work has focused on changes of specific glycans on specific proteins, and demonstrating functional

consequences of these glycosylation changes on cancer growth, immune function, and metastasis (24-27).

Recent advances in mass spectrometry (MS)-based methods have significantly expanded the capabilities for quantitative glycoproteomics, enabling highly sensitive and accurate quantitation of glycosylation at intact glycopeptide level (28, 29). What has emerged from these studies is a complex array of glycosylation changes that occur in many malignancies. For example, in prostate cancer, one common finding is an increase in the number and type of glycans at specific glycosites in cancer compared to normal tissues (30). Looked at another way, in prostate cancer, glycan diversity is increased at specific glycosites when compared to non-cancerous prostate tissues. Although systematic analysis of glycoproteins in preclinical and clinical studies has revealed that aberrant glycosylation is closely associated with RCC, whether the glycosylation diversity is significantly different between renal tumors and adjacent normal tissue is not clear.

One significant challenge arising from high-throughput glycoproteomics analysis is capturing and analyzing these changes in glycan diversity. To overcome this challenge, we employed the Shannon diversity index, a commonly used metric in ecology for measuring the biodiversity of a community (31) that has also been applied in molecular biology (32). Specifically, we used this metric to capture the degree of glycosylation diversity across all identified glycoproteins (glycoproteome), within individual glycoproteins (meta-heterogeneity), and at specific glycosites on proteins (micro-heterogeneity) (33-35). We used this systematic evaluation of glycosylation diversity in surgically resected renal tumors and paired adjacent normal kidney tissues to identify proteins, glycosites, domains, and pathways in which tumor-associated diversity changes were concentrated. By integrating these diversity measurements with total-protein abundance and RNA-sequencing (RNA-seq), we sought to distinguish abundance-associated

changes from site-specific glycosylation patterns and to nominate candidate transcript-glycosite associations.

METHODS

Experimental Design and Statistical Rationale

Tissue Samples

Paired normal (N = 32) and renal tumor (N = 32) samples were harvested from patients (N = 32) undergoing radical nephrectomy for presumed malignancy based on clinical and imaging characteristics. Each normal tissue represented a patient-matched control, for a total of N = 32 normal tissue biological replicates and N = 32 renal tumor biological replicates. Sample size was determined by the availability of paired frozen renal tumor and adjacent normal tissues that met the study inclusion criteria, rather than by an a priori power calculation. The paired design was used to control for inter-patient biological variation. All patients had signed informed consent for the use of their tissues under an Institutional Review Board- approved protocol (IRB#: 13828). Final pathology characterized tumors histologically as clear cell renal cell carcinoma (n=24), papillary renal cell carcinoma (n=1), chromophobe renal cell carcinoma (n=1), oncocytoma (n=4), transitional cell carcinoma (n=1), and angiomyolipoma (n=1) (Supplementary Table 1). Samples were harvested and immediately frozen in liquid nitrogen and stored at -80 °C until analysis. Tumor samples were mounted on chucks with OCT (optimal cutting temperature) embedding medium, followed by frozen sectioning and hematoxylin and eosin staining. Tumor specimens were confirmed histologically to contain >90% viable tumor without necrosis or visible contamination by normal kidney, and adjacent-normal specimens were confirmed to be free of

tumor and inflammatory contamination. All visible OCT medium was trimmed from the samples prior to sample processing for analysis.

Trypsin Digestion and Glycopeptide Enrichment

To reduce potential batch and order effects, paired tumor and adjacent-normal samples were randomized across sample-preparation batches before digestion and glycopeptide enrichment.

Renal cell carcinoma tissue samples were mechanically homogenized using a PRO-250 homogenizer in 1 mL of lysis buffer consisting of 1.5% sodium dodecyl sulfate (Thermo Fisher Scientific) and 1X protease inhibitor (Sigma-Aldrich), followed by ultrasonication with a Branson probe sonicator (Fisher Scientific) for three 15 s cycles with the amplitude set at 40%. Protein concentration was quantified using a standard BCA protein assay (Thermo Fisher Scientific) following the manufacturer's protocol. Three hundred micrograms of protein from the tissue lysates were aliquoted, and the sample volumes were brought to 80 µL using 50 mM ammonium bicarbonate (Sigma Aldrich). Samples were then vortexed for 15 s and incubated for 1 hour at 65 °C with 10 mM Tris(2-carboxyethyl) phosphine hydrochloride (TCEP, Sigma Aldrich) to reduce cystine disulfide bonds. Free thiol groups on cysteines were then alkylated with 15 mM of iodoacetamide (Across Organics) in the dark for 45 min at room temperature. Proteins were precipitated with 1 mL of cold acetone and stored at -20 °C overnight. The precipitated proteins were pelleted via centrifugation at 14,000 x *g* for 10 mins at 4 °C, reconstituted with 80 µL of 50 mM ammonium bicarbonate, digested with high-grade sequencing trypsin enzyme (Thermo Fisher Scientific) in a 1:25 enzyme-to-protein ratio, and incubated at 37 °C overnight. Glycopeptides were enriched using strong anion exchange and electrostatic repulsion hydrophilic interaction

chromatography (SAX-ERLIC) as previously described (36). Briefly, the SOLA SAX SPE (Thermo Fisher Scientific) columns were washed with 3 mL of acetonitrile followed by 3 mL of 100 mM triethylammonium acetate in water, and 3 mL of 1% trifluoroacetic acid (TFA) in water. Columns were equilibrated with 3 mL of equilibration solution consisting of 1% TFA, 95% acetonitrile, and 4% water. Next, 3 mL of equilibration solution was added to each sample, which was then loaded onto the SOLA-SAX SPE columns. The SOLA-SAX SPE columns were washed with 6 mL of equilibration solution. Glycopeptides were eluted by adding 850 µL of 0.1% TFA in 50% acetonitrile, and 49.9% water twice, followed by another two 850 µL aliquots of 0.1% TFA, 5% acetonitrile, and 94.9% water. The two glycopeptide fractions were concentrated, combined, and dried down using a speed vacuum (LabConco). Enriched glycopeptides were further fractionated using high-pH reversed-phase spin columns (Thermo Fisher Scientific) following the manufacturer's protocol.

Glycopeptide LC-MS/MS Analysis

LC-MS/MS of glycopeptide samples was performed on an Orbitrap Eclipse Tribrid mass spectrometer (Thermo Fisher Scientific) coupled with a Dionex UltiMate 3000 RSLCnano UHPLC system (Thermo Fisher Scientific). Samples were randomized for analysis order, and a standard bovine serum albumin (BSA) digest was run daily to monitor instrument performance. Glycopeptide fractions were reconstituted in 12 µL of 0.1% formic acid in water and loaded on a C18 PepMap trap column (Thermo Fisher Scientific) at a flow rate of 5 µL/min for 10 min. Glycopeptides were separated by reversed-phase chromatography on a 25 cm long C18 analytical column (New Objective) packed in-house with BEH C18, 130 Å, 1.7 µm particle size resin (Waters) encapsulated in a column heater (MSWIL) set to 60 °C. Glycopeptides were eluted at a

flow rate of 0.3 µL/min by changing the mixture of mobile phase A (0.1% formic acid in water) and mobile phase B (0.1% formic acid in acetonitrile) as follows: mobile phase B was held at 2% for the first 6 min, slowly ramped up to 35% over the next 80 min, followed by an increase to 85% over 5 min with a 5 min hold before re-equilibrating the analytical column with 2% mobile phase B for 10 min before the next sample injection. Blank injections were included between samples to monitor and reduce carryover. Enriched glycopeptide fractions were injected in reverse order of expected glycopeptide concentration, beginning with the lowest-concentration fraction and proceeding sequentially to the highest.

Samples were subjected to electrospray ionization with an ionization voltage of 1.8 kV in positive mode using a nanospray Flex ion source (Thermo Fisher Scientific) coupled to the Orbitrap Eclipse system. The MS parameters used for MS/MS data-dependent acquisition (DDA) consisted of setting the cycle time at top-speed for 3 s with a scan mass range set to 375-2,000 *m/z*, an Orbitrap mass resolution of 120 K at *m/z* 200, the normalized automatic gain control (AGC) target set to 250%, and maximum injection time mode set to auto for the MS1 precursor ions. The most abundant precursor ions were fragmented using high-energy collision-induced dissociation (HCD) with a collision energy of 38%. Dynamic exclusion was enabled for 15 s with a mass tolerance of 10 ppm. The quadrupole isolation window was set to 1.2 *m/z*, and the normalized AGC target was set to 200%. The MS2 fragmented ions were detected in the Orbitrap with a mass resolution 30 K at *m/z* 200. The raw data files were searched using Byonic v3.9.6 (Protein Metrics) software against a focused protein database generated from the shotgun proteomic analysis from the same renal tumors sample set (2020; 5,762 entries, see next section) using Uniprot retrieve/ID mapping tool and the Byonic library of 309 mammalian N-glycans. The glycopeptide search parameters included: trypsin digestion (fully specific) with a maximum of two missed cleavages,

MS1 precursor mass tolerance of 10 ppm, and fragment mass tolerance of 0.5 Da with fixed cysteine carbamidomethylation and variable methionine oxidation and asparagine deamination modifications. A false discovery rate (calculated by Byonic) filter of < 1% was applied to protein identifications.

LC-MS/MS Proteomic Analysis

Twenty-five micrograms of protein from each sample were digested with trypsin and analyzed similarly to the glycopeptide-enriched samples described above, again using a standard BSA digest to monitor instrument performance. Samples were randomized for analysis order, and each sample was injected in triplicate to maximize protein identifications and to ensure instrument precision. Briefly, samples were analyzed by loading 1.5 µg of the tryptic peptides on a C18 PepMap trap column using a nanospray Dionex Ultimate 3000 RSLCnano UHPLC system (Thermo Fisher Scientific) in-tandem to an Orbitrap Elite (Thermo Fisher Scientific) hybrid mass spectrometer at a flow rate of 5 µL/min for 10 min. A 25 cm long C18 analytical column (New Objective) packed with Magic C18 AQ, 100 Å, and 5.0 µm resin was used to chromatographically separate tryptic peptides using a 0.5 µl/min flow rate. Mobile phase B was held at 2% for the first 10 min, slowly ramped up to 35% over the next 110 min, followed by an increase to 85% over 5 min with a 5 min hold, and an equilibration time of 10 min with mobile phase B. The top 10 most abundant ions per MS1 scan were selected for high-energy collision-induced dissociation (35 eV) in a data-dependent manner. MS1 mass resolution and the FT AGC were set to 60 K at *m/z* 200 and 1e6, respectively. The *m/z* scan range was set from *m/z* 400-1,800 with a MS2 AGC target at 3e4. Dynamic exclusion was enabled for 30 s. Raw data files were searched using Byonic v3.9.6 software against a Swiss-Prot database reference human proteome (2020; 20,366 entries),

including search parameters of trypsin digestion (fully specific) with maximum of two missed cleavages, MS1 precursor mass tolerance of 10 ppm, and fragment mass tolerance 0.5 Da with fixed cysteine carbamidomethylation and variable methionine oxidation and asparagine deamination modifications. A false discovery rate (calculated by Byonic) filter of < 1% was applied to protein identifications.

Data Processing and Analysis

Intact glycoproteomics searches were performed using Byonic with decoy generation enabled, and FDRs were estimated using a target-decoy strategy implemented in Byonic. Glycopeptide identifications were retained for downstream analysis if they passed both the protein-level FDR threshold (< 1%), and the glycopeptide-level confidence filters (Byonic Score > 150 and log probability > 1.5), which were selected to restrict the dataset to high-confidence glycopeptide spectrum matches for site-specific glycosylation analysis. To avoid ambiguous glycan assignments in downstream analysis, Byonic glycopeptide-spectrum matches with more than one glycan composition assigned to the same peptide-spectrum match were excluded before glycopeptide filtering and quantification. Glycopeptide identifications were corrected for misidentified monoisotopic masses due to the 1 Da difference between two sialic acids and one fucose (Fuc) as described previously (37). One sialic acid (NeuAc) was substituted for two fucoses per -1 Da off-by-x error. Next, each corrected putative glycan structure was classified into one of seven categories including: incomplete core, high mannose, hybrid, complex undecorated, complex sialyated, complex fucosylated, and complex fucosylated + sialyated, using the criteria described below. All glycans containing ≤ 2 N-acetylhexosamine (HexNAc) and ≤ 3 hexose (Hex) were classified as incomplete core. Glycans with HexNAc = 2 and Hex > 3 were classified as high

mannose. Glycans with HexNAc ≥ 2, Hex ≥ 3, and Hex outside the chitobiose core < HexNAc(s) were classified as hybrid. Glycans that were not considered incomplete core, high mannose, or hybrid were classified as complex undecorated (no Fuc or NeuAc), complex sialylated (NeuAc ≥ 1, Fuc = 0), complex fucosylated (Fuc ≥ 1, NeuAc = 0), or complex fucosylated + siaylated (NeuAc ≥ 1, Fuc ≥ 1). The specific position of the asparagine in a protein modified by each glycan (i.e., glycosites) was determined by searching a human proteome database in conjunction with the glycan database as described above. For N-linked glycopeptide searches, Byonic assigns glycans to asparagine residues within N-glycosylation sequons using the N-glycan search setting, peptide sequence information, precursor mass, fragment ion evidence, and the specified glycan database. Because N-glycan site assignment was constrained by the presence of an N-glycosylation consensus sequon (N-X-S/T, where X is not proline), no separate glycosite-localization score threshold was applied. Each glycosite was mapped to functional domains according to PROSITE (38). Protein-level accessions, protein groups, unique peptide counts, shared peptide information, and protein sequence coverage were taken from the Byonic protein output tables generated for each search.

Filtered glycopeptide-spectrum matches are provided in Supplementary Table S2. For each identification, the table reports peptide sequence, protein name and database accession, glycan assignment and monosaccharide composition, observed modification type(s), precursor charge, observed precursor m/z, observed and calculated protonated mass, mass error, peptide and glycosite position, scan number, retention time, raw-file and sample identifiers, Byonic Score, Delta, Delta Mod, absolute Log Probability, and the broad glycan category used in the diversity analyses.

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The heterogeneity of each identified glycoprotein and glycosite were quantified using Shannon diversity index:

$$H' = -\sum p_i \cdot \ln(p_i)$$

Where p is the frequency (n/N) of individuals of one particular species found (n) divided by the total number of individuals found (N) and i is the number of species found (39). In this study, the species are the glycan categories considered in this analysis: incomplete core, high mannose, hybrid, complex undecorated, complex sialyated, complex fucosylated, and complex fucosylated + sialyated, and the census metric used is the counting of glycopeptide spectrum matches identified at the glycoproteome, glycoprotein, or glycosite level.

Differences in Shannon diversity between two groups were evaluated using non-parametric Mann-Whitney U tests because glycopeptide spectrum-match counts and Shannon diversity values were not assumed to follow a normal distribution. For protein-, glycosite-, and transcript-level families of comparisons, p-values were adjusted using the Benjamini-Hochberg procedure to control the false discovery rate; a Benjamini-Hochberg-adjusted p-value < 0.01 was considered significant unless otherwise specified. Raw p-values were retained only for explicitly labeled exploratory ranking or visualization. Z-tests were used to assess whether Shannon diversity values differed from the population-level distribution. WebGestalt (40) overrepresentation analysis of Gene Ontology Biological Process terms was restricted to categories with more than five proteins, raw $p < 0.001$, and enrichment > 4.

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE (41) partner repository with the dataset identifier PXD075353. The PRIDE submission contains 512 raw mass spectrometry files. Two deposited raw acquisitions, RCC_2B_Fxn2.raw and RCC_2B_Fxn3.raw, were incomplete or corrupted and could not be

processed; consequently, no corresponding Byonic .byrslt, .byspec2, or exported .xlsx result files were generated, and these acquisitions were excluded from downstream analyses. For RCC_54B_Fxn6_Rerun.raw, the .byspec2 file and exported .xlsx results are available in PRIDE, whereas the native .byrslt file is not available. All other submitted .byrslt and .byspec2 files were verified in Byonic Viewer.

RNA Isolation

Total RNA was isolated from frozen from normal and tumorous kidney tissues using TRIzol™ reagent (Invitrogen, catalog no. 15596026) according to the manufacturer's instructions. Specifically, 50-100 mg of frozen tissue was minced, suspended in 1 mL of TRIzol™, and homogenized using a handheld Tissue-Teror homogenizer (Biospec Products, Inc.). After resting for 5 min, 0.2 mL of chloroform was added, and each sample was gently shaken for 2–3 minutes. The samples were centrifuged for 15 minutes at 12,000 × *g* at 4°C. The aqueous phase containing the RNA was transferred to a new tube, and 0.5 mL of isopropanol was added and incubated for 10 min at room temperature. The sample was centrifuged for 10 min at 12,000 × *g* at 4°C, and the supernatant was discarded. The pellet was washed with 75% ethanol and air dried for 10 min. The pellet containing RNA was resuspended in RNase-free water (50 µL) and incubated in a water bath at 55°C for 10 min to dissolve the RNA.

RNA Sequencing and Analysis

The sequencing library was prepared using SMARTer® Stranded Total RNA-Seq Kit v3 (catalog. 634486, Takara Bio USA, Inc) with service provided by Novogene. Briefly, total RNA was converted to cDNA, and then Illumina adapters and barcodes were added through PCR (5

cycles of 98 ºC for 15 s, 55 ºC for 15 s, 68 º for 30 s). PCR products were purified with NucleoMag beads, and ribosomal cDNA was subsequently depleted. The resulting products were further amplified with universal primers and purified with NucleoMag beads to obtain final cDNA libraries for sequencing. The prepared libraries were sequenced on an Illumina NovaSeq 6000 using paired-end (PE) mode with a read length of 150 base pairs (2×150 bp). Each sample was sequenced to a depth of approximately 50 million read pairs, yielding about 15 billion read pairs per sample.

Raw sequencing reads were aligned to the Ensembl GRCh38 reference genome using STAR (42) with the following parameters: outFilterScoreMinOverLread=0, outFilterMatchNminOverLread=0, outFilterMatchNmin=0, and outFilterMismatchNmax=2. Gene counts were calculated using HTSeq (43). FastQC was used to assess the quality of the sequencing data, which was subsequently aggregated with MultiQC,(44) along with STAR alignment scores and HTseq counts. All proceeding steps were managed using Snakemake (45). Transcript counts were then normalized to the total of all non-mitochondrial transcripts detected in each sample (median 15% of all transcripts were of mitochondrial origin), of which typically 50-70% were protein-coding genes. Gene expression data have been deposited in Gene Expression Omnibus (GEO accession no. GSE325575).

Omics Data Normalization

To facilitate comparisons across datasets, proteomic and transcriptomic data were normalized using similar methods. For the proteomics data, protein intensities were first computed by summing the intensities of all peptides assigned to a given protein group. Log_2 ratios were then calculated for the intensity of each protein in a given sample divided by the average intensity of

that protein across all samples in which it was detected. To correct for global shifts and other technical variation across injections, $\log_2$ ratios were normalized within each sample by subtracting the sample-specific mean and dividing by the standard deviation, resulting in a normalized distribution centered at zero. For cross-platform analyses, proteins were retained only when identified with more than one peptide-spectrum match and quantified in at least 40% of the samples.

For RNA-seq data, gene abundances were calculated as transcripts per million (TPM). For each gene, the $\log_2$ ratio between the TPM in each sample divided by the average TPM of that gene across all samples was calculated. Log_2-ratio values were then normalized using the same z-score approach used with the proteomics analysis described above, yielding consistent scaling across datasets and allowing downstream integrative modeling.

Protein Abundance and Glycosylation Diversity Association Analysis

To evaluate total protein abundance as a potential confounder, tumor-versus-adjacent-normal changes in normalized protein abundance were compared with the corresponding changes in Shannon diversity at the protein and glycosite levels. Analyses were restricted to glycosylation features quantified in at least 40% of samples and with matched total-protein measurements. Pearson correlations and univariable linear regressions were calculated, and R2 was used to summarize the proportion of linear variation in the diversity change associated with the protein-abundance change. These analyses quantify association and do not establish causality.

Feature Selection and Integrative Modeling of Candidate Glycosylation Associations

To investigate candidate molecular associations with the observed site-specific heterogeneity, we employed a two-step strategy combining statistical testing and penalized regression modeling. First, we identified glycosites with significantly increased heterogeneity in tumor samples by performing a Mann-Whitney U test on the Shannon diversity index. Glycosites showing a p-value < 0.05 and a positive difference in Shannon index between cancer and non-cancerous samples were used as input of the next step.

For each glycosite, we calculated the individual components of the Shannon index per glycan type, i.e., $-p_i \cdot \ln(p_i)$ and used these as predictors in a uniLASSO logistic regression model (46). We applied a penalized regression analysis using uniLASSO to select the most relevant variables associated with the observed phenotype (increased glycosite heterogeneity in cancer). The model was trained using a 70:30 training-to-testing split and was repeated 10,000 times. Glycan-type features and their corresponding glycosites were recorded if they were selected in each model iteration.

In parallel, we applied the same feature-selection strategy to the RNA-seq data using normalized expression levels of the 66 glycosyltransferase or glycan-processing genes listed in Supplementary Table S5 as candidate predictors (47)Only enzyme transcripts with significant upregulation in the tumors compared to non-tumor samples (p-value < 0.05 and positive difference in expression in the tumors) were eligible for the uniLASSO models. Model selection was used to nominate associations and was not interpreted as evidence that a selected transcript directly caused a glycosylation change.

To identify and visualize plausible relationships between glycan categories and glycosylation-related transcripts, we integrated the results from both uniLASSO analyses. Published literature was reviewed to assess the biological plausibility of each selected association;

this review did not constitute experimental validation or direct assignment of an enzyme to a specific glycan structure.

To increase clarity and avoid over-saturation in the final representation, we included only glycan-type–glycosite pairs selected in > 3% of the model iterations and enzyme transcripts selected in > 1% of the model iterations. The filtered relationships were visualized using a Sankey diagram, where nodes represented either enzymes or glycan-type–glycosite pairs, and the width of the connection indicated the proportion of models in which each feature was selected. The diagram is a hypothesis-generating summary of model-selected associations, not a definitive biosynthetic pathway.

RESULTS

Individual Glycoproteins from Renal Tumor Tissues Exhibited Significantly Higher Levels of Glycosylation Heterogeneity Compared to Adjacent Normal Tissues

To characterize systematically the glycoproteomic landscape of renal tumors and explore underlying differences compared to adjacent normal kidney tissues, we conducted a glycoproteomic analysis of surgically removed tumor and adjacent normal tissue samples. We performed LC-MS/MS analysis of intact glycopeptides from 32 pairs of tumor and normal tissue samples from males and females, 39 to 82 years old, with detailed clinical and histological characterization (Supplementary Table S1). We identified 853,340 total and 54,354 unique N-linked glycopeptides from 1,940 glycoproteins containing 220 glycan compositions. Each glycopeptide was categorized by glycan type (i.e. incomplete core, high mannose, hybrid, complex undecorated, complex sialylated, complex fucosylated, or complex fucosylated + sialylated). In tumor tissue, complex sialylated (Sia) glycans were the most abundant putative glycan structures,

whereas in adjacent normal tissue, complex fucosylated + sialylated (Fuc + Sia) glycans were the most abundant (Figure 1A). Filtered identifications and their confidence metrics are provided in Supplementary Table S2.

To quantify the extent of protein glycosylation heterogeneity in renal tissue samples, we calculated Shannon diversity index values, incorporating information about glycan categories and the distribution of glycopeptide spectrum matches (GSMs) among different glycan categories. The Shannon diversity index values were then used to compare the degree of glycosylation heterogeneity systematically across all identified glycoproteins (glycoproteome), within individual glycoproteins (meta-heterogeneity), and at specific glycosites on proteins (micro-heterogeneity). Comparison of global glycosylation heterogeneity across all N-glycosylation sites at the glycoproteome level showed no significant differences between renal tumors and adjacent non-cancerous tissue (Figure 1B). Although there were more glycopeptides identified in the tumor tissues compared to the adjacent normal tissue, the distribution of glycans across categories, as calculated by Shannon diversity index, was not significantly different. Because the Shannon index is calculated from relative glycan-category proportions, it is not directly determined by absolute glycopeptide spectrum-match abundance; however, data-dependent acquisition and sampling depth may still affect detection of low-abundance glycoforms.

To determine local glycosylation heterogeneity at the specific protein level, we calculated Shannon diversity index values for individual glycoproteins with at least two identified glycopeptides. Overall, individual glycoproteins from renal tumor tissues exhibited significantly higher levels of glycosylation heterogeneity compared to adjacent normal tissue ($P < 0.0001$) (Figure 1C).

Glycoproteins with Increased Glycosylation Diversity in Renal Tumors are Involved in Vital Cancer-Relevant Biological Processes

For an exploratory view of the proteins with the largest group differences, we ranked glycoproteins by the Mann-Whitney U raw p-value and clustered the 100 proteins with raw $p < 0.01$ (Figure 2A). Raw p-values were used to rank features for this display, whereas Benjamini-Hochberg-adjusted p-values were used for multiple-testing inference. Of these proteins, 86 showed increased glycosylation heterogeneity in renal tumors compared to adjacent normal tissue (top of heatmap), while only 14 proteins showed decreased glycosylation diversity (bottom of heatmap). To better understand the biological functions of glycoproteins with higher glycosylation diversity in renal tumors, we performed an overrepresentation analysis of the 86 proteins using the Gene Ontology Biological Process database (Figure 2B). The most significantly enriched category, containing the largest number of proteins, was extracellular structure organization. The remainder of the significantly enriched processes included categories related to the immune response (neutrophil mediated immunity, granulocyte activation, acute inflammatory response, tumor necrosis superfamily cytokine production, phagocytosis, and leukocyte migration), and cell adhesion (leukocyte cell-cell adhesion, cell-substrate adhesion, heterotypic cell-cell adhesion, and integrin-mediated signaling pathway).

Descriptive Glycan-Category Shifts in Renal Tumors

The three largest glycoprotein clusters (#3, #5, and #6) that showed increased glycosylation heterogeneity in tumor tissue accounted for ~63% of all significant proteins (Figure 2C, Figure S2). The largest cluster (#3), containing 24% of the significant glycoproteins, was predominantly decorated with high mannose glycans, and showed more than a two-fold increase in all glycan

subtypes except for a decrease in high mannose in renal tumors compared to adjacent normal tissues (Figure S3). The second largest cluster (#6) contained 23% of significant glycoproteins and was predominantly decorated with complex sialylated + fucosylated glycans in renal tumors compared to adjacent normal tissues. The third largest cluster (#5) was predominantly decorated with complex sialylated glycans and showed a significant increase in glycans with complex fucosylation, fucosylation + sialylation, and incomplete core, and a decrease in complex fucosylated glycans in kidney tumors compared to adjacent normal tissues. While the fucosylation increase is notable, these category-level shifts are descriptive and do not by themselves establish that a specific glycosyltransferase, glycosidase, or biosynthetic pathway caused the observed pattern.

Protein Glycosylation Differences Commonly Observed Between Renal Tumors and Adjacent Normal Tissues

To identify commonly observed glycosylation changes, we focused on 177 proteins for which a diversity index could be calculated in at least 40% of the samples. Tumor-versus-adjacent-normal comparisons were evaluated with Mann-Whitney U tests and Benjamini-Hochberg correction (Figure 3A). The direction of the largest differences was generally consistent across biological samples, supporting the internal robustness of the tumor-associated patterns rather than establishing a universal signature.

Proteins with the most pronounced differences based on statistical significance and effect size are shown in a volcano plot (Figure 3B). Specifically, fibronectin (FN1) and MAN2B1, essential proteins in extracellular matrix remodeling and glycan catabolism, showed consistent and robust increases in protein expression levels and glycosylation heterogeneity in the tumors. In

addition, IGHA2, AOC3, COL6A1, POSTN, A2M, and clusterin (CLU), which have been previously associated with immune modulation, angiogenesis, and tumor cell plasticity, showed a similar pattern of increased expression and glycan diversity. Increased levels of fucosylation and siaylation of CLU at N374 have been reported in the plasma of clear cell renal cell carcinoma patients (48). Our data showed significantly increased overall protein levels and increased glycosylation heterogeneity of CLU in renal tumors, including increased fucosylation and sialylation at N374 (Figure S4). In contrast, several proteins, including MMRN2, PLXNB2, and TMED9, displayed increased glycosylation diversity despite decreases in overall protein levels in the tumors, indicating that protein abundance alone does not determine glycosylation diversity. In the renal tumor samples, three proteins, CLPTM1L, UGT1A9, and AHSG (Figure 3B), showed a significant correlation between glycosylation diversity index and cancer stage (Pearson $P < 0.01$).

Protein Abundance Was Associated with but Did Not Fully Account for Glycosylation Diversity

Across matched features, tumor-versus-adjacent-normal changes in total-protein abundance were moderately positively associated with changes in Shannon diversity at both the protein (R2 = 0.290) and glycosite (R2 = 0.279) levels (Figure S1A-B). Thus, protein abundance accounted for approximately 29.0% and 27.9% of the linear variation in the corresponding diversity changes, whereas 71.0% and 72.1%, respectively, were not accounted for by these univariable linear models. The associated feature-level statistics are provided in Supplementary Tables S3 and S4.

Several proteins showed higher glycosylation diversity despite lower abundance, and different glycosites within the same protein could change in opposite directions. For example, mean diversity at LAMP1 N249 was higher in adjacent normal tissue, whereas mean diversity at LAMP1

N261 was higher in tumor tissue; only the N249 comparison remained significant after Benjamini-Hochberg correction (Supplementary Table S3). These results support a site-specific, multifactorial interpretation and show that higher diversity cannot be attributed solely to higher protein abundance or deeper sampling.

Glycosylation Differences Commonly Observed between Renal Tumors and Adjacent Normal Tissues are Site-Specific

To better understand whether changes in glycosylation occurred at multiple or single sites on proteins, we examined glycosylation changes at specific glycosites. Several proteins showed concordant increases in glycosylation diversity across multiple sites in renal tumors compared to adjacent normal tissue. For example, FN1 displayed multiple significantly altered glycosites (Figure 3C) across eight FN1 glycosites (N430, N528, N542, N877, N1007, N1236, N1291, and N2199) in at least four patients (Figure 4). Four of these sites (N430, N528, N542, and N1007) showed statistically significant increases in glycosylation diversity in renal tumors compared to adjacent normal tissue. N430 (complex sialylated, complex undecorated, high mannose, and hybrid), N528 (complex sialylated and hybrid), and N542 (complex fucosylated, complex sialylated, high mannose, and hybrid) are located within the collagen-binding domain of FN1. N1007 (complex fucosylated and complex sialylated) is located within the central region of fibronectin, near the boundary of type III fibronectin repeats, where it contributes to FN1's modular flexibility.

Other proteins showed glycosylation differences driven by specific glycosites. For instance, CLPTM1L possessed a predominantly high mannose glycosylation profile in adjacent

normal kidney tissue, but showed striking diversity in the cancer tissues, with consistent increases in complex fucosylation and some increase in hybrid glycan structures (Figure S5).

Interestingly, the glycosylation differences were primarily driven by a single glycosite at N91 (Figure 3C, Figure S6, Figure S7), found in the large extracellular domain of CLPTM1L. Similarly, UGT1A9 showed dominantly high mannose glycans in normal tissue, whereas complex fucosylated glycans were significantly ($P < 0.0001$) increased in cancer (Figure 5A). This increase was predominantly driven by two glycosites, N71 and N344, which showed a significant correlation with cancer stage (Pearson $p < 0.01$) in renal tumors and adjacent normal tissues. The heatmap in Figure 5B shows the distribution in kidney tumors and normal tissue of glycosylation types for two glycosites of UGT1A9 (N71 and N344) for each patient. The complex fucosylated glycan structures at N71 and N344 in renal tumors and adjacent normal tissues are shown in Figure 5C.

We also identified cases where the overall protein-level glycosylation diversity remained unchanged, but individual glycosites displayed significant differences between tumor and normal tissues. To better understand this observation, we searched for proteins with no significant differences in glycosylation diversity at the overall protein level ($p > 0.1$) with at least three characterized glycosites. We found 13 glycoproteins with one or more glycosites whose diversity was significantly altered in tumors compared to normal, including AZGP1, COL12A1, F2, HSPG2, HYOU1, ITGA1, ITGA3, ITGAV, LAMA5, ORM1, SERPINA3, SERPING1, and VCAN. These results suggest that glycosylation differences commonly observed between renal tumors and adjacent normal tissues are site-specific.

Differences in Site-Specific Glycosylation are Linked to Particular Protein Domains

We next examined whether the glycosylation diversity observed at the glycosite level was associated with structural and functional domains of the 13 proteins. We first mapped the glycosites showing high diversity to protein domains using Prosite (38). We observed a statistically significant increase ($p = 0.008$) in glycosylation diversity in intradomain compared to interdomain regions in renal tumors and a similar trend in normal tissue, albeit less significant (Figure S8A), suggesting a potential role of these modifications in protein functions. Within intradomain regions, glycosites in complex but not simple interactor domains showed increased glycosylation diversity compared to interdomain regions in renal tumors, while in adjacent normal tissue, glycosites in simple but not complex interactor domains showed increased glycosylation diversity (Figure S8B). Finally, in both tumor and normal tissue, glycosites in LAMP domains showed increased glycosylation diversity compared to interdomain regions. These observations hint at the intriguing possibility that site-specific changes in diversity could have functional implications in renal tumors.

Transcript Levels of Glycosylation-Related Enzymes are Altered in Renal Tumors

Given the increase in complex fucosylation observed in renal tumors, we evaluated transcript levels of glycosylation-related genes using bulk RNA-seq (49). Among the 66 glycosyltransferase or glycan-processing genes listed in Supplementary Table S5, 22 showed differential expression at a Benjamini-Hochberg-adjusted p-value < 0.01; all but FUT6 showed higher expression in tumors. Within this set, FUT6 was significantly decreased and FUT8 was significantly increased in tumors (Figure 6A). GFUS was also significantly increased and was retained in Figure 6A because it encodes GDP-L-fucose synthase and is relevant to fucosylation through GDP-fucose biosynthesis; however, it is not a fucosyltransferase and was not included

among the 66 genes used in the downstream RNA-seq integration analyses. FUT11, also known as POFUT4, was excluded from the N-glycan pathway interpretation because it is a protein O-fucosyltransferase and cannot be assigned to the N-glycan categories measured here. Among the ST3GAL genes, ST3GAL2 rather than ST3GAL1 showed a significant tumor-associated increase; ST3GAL1 showed only a small, non-significant change (Figure 6C). The enzymes shown in Figure 6B represent plausible linkages rather than causal assignments to the measured glycan categories.

Finally, glycosyltransferases and glycosidases involved in glycan addition, trimming, remodeling, and extension showed altered RNA expression in renal tumors compared with adjacent normal tissue (Figure S9). Collectively, these results nominate transcriptional changes that may contribute to the heterogeneous glycosylation landscape but do not establish direct enzymatic causality.

Many glycosyltransferases and glycosidases were not consistently detected in the unenriched shotgun proteomics dataset, which is expected for low-abundance ER/Golgi and membrane-associated enzymes. After requiring more than one peptide-spectrum match and quantification in at least 40% of samples, GANAB was the only glycosylation-related protein retained. RNA-seq was therefore used to provide broader pathway coverage, whereas total proteomics was used to evaluate protein abundance as a potential confounder of glycosylation diversity (Figure S9; Supplementary Tables S4 and S5).

Integrative Modeling Identified Candidate Enzyme-Glycosite Associations

To systematically investigate the potential associations between altered expression of glycosylation enzymes and specific glycosylation changes observed in renal tumors, we explored an integrative modeling approach combining transcriptomic and glycoproteomic data. We first

identified glycoproteins and glycosites with the most significant increases in glycosylation diversity in renal tumors using a uniLASSO-based feature selection analysis (46). The most frequently selected glycosite–glycan-type combinations are shown in Figure S10. Among the top features, THBS1 (N1067, hybrid) and FN1 (N542, complex fucosylated and hybrid) were consistently selected across more than 5,000 models, indicating their frequent model selection in tumor-associated glycosylation shifts. Additional frequently selected sites included PSAP (N332, complex fucosylated + sialylated), ACAN (N333, complex undecorated), HSP90B1 (N217, complex sialylated), APOD (N98, complex undecorated), and APMAP (N160, both high mannose and complex fucosylated), each reflecting diverse changes in glycosylation patterns in renal tumors compared to normal tissue. To nominate candidate transcript-level associations with the observed glycosylation patterns, we integrated these results with a parallel uniLASSO model based on RNA-seq data of glycosylation enzymes. The most frequently selected genes included MGAT1, DPAGT1, B3GNT8, ST3GAL3, and ST3GAL5, which were reported as candidate predictors rather than definitive enzymatic drivers of the observed glycosylation patterns (Figure S11) (50).

We constructed a Sankey diagram displaying the model-selected enzyme-transcript-glycosite-glycan-category associations (Figure 7). Only glycan-type–glycosite pairs selected in >3% of the models and enzymes selected in >1% of the models were included. The width of each connection corresponds to the frequency of co-selection across all model iterations. The diagram is intended as a hypothesis-generating visualization rather than a definitive biosynthetic pathway. It highlights candidate associations that can prioritize future targeted experiments but does not establish that increased transcript expression causes increased glycosylation diversity.

DISCUSSION

High-throughput MS-based determination of intact glycosylation reveals heterogeneous glycans at the glycopeptide level across many protein glycosites in kidney tumors and paired normal tissues, with the largest tumor-associated differences concentrated at selected proteins and glycosites rather than at the whole-glycoproteome level. Given this complexity, a significant challenge lies in identifying which of the many glycosylation changes are biologically important in promoting tumor growth. The changes between kidney tumors and normal tissues do not simply reflect glycosylation infidelity in the tumors, since global glycosylation diversity is similar between the two tissue types. Rather, glycan site heterogeneity, measured using the Shannon diversity index, shows discrete, site-specific changes in tumors compared with paired adjacent normal kidney tissues. These patterns should be interpreted as reproducible, hypothesis-generating associations rather than direct evidence that a particular glycosylation change promotes tumor growth.

Enrichment analysis demonstrates that differences in diversity between renal tumors and normal tissue are significantly enriched for extracellular matrix proteins. This observation agrees with an analysis of total glycopeptide differences between clear cell RCC and paired normal adjacent tissues, demonstrating enrichment for extracellular matrix (ECM) receptor interaction, cell adhesion molecules, and focal adhesion molecules (51). In addition, our enrichment analysis pinpointed significant changes in glycoprotein diversity related to immune function. A large body of data demonstrates the importance of glycosylation in immune function and immune evasion in many malignancies. Our observation that renal tumors are enriched for complex sialylation is highly relevant, as abundant data demonstrate that sialylated glycoproteins can serve as ligands for sialic acid-binding immunoglobulin-type lectin (Siglec) receptors on immune cells, acting as immune checkpoints across many malignancies (52).

Previous renal cancer glycomic and glycoproteomic studies provide important context. In a tandem mass tag (TMT)-based analysis of 103 clear cell renal cell carcinoma tumors and 80 paired adjacent normal tissues, high-mannose (44.8%) and sialylated (32.5%) species accounted for substantial proportions of tumor-upregulated intact glycopeptides, whereas fucosylated species accounted for 41.6% of those downregulated; consensus clustering also identified three glycoproteomic tumor subtypes (51). Pooled-plasma discovery studies before and after curative nephrectomy reported alterations in high-mannose, sialylated, and fucosylated N-glycans, including site-specific changes in clusterin glycoforms at N374 (48, 53). More recently, lectin-microarray profiling of 50 paired clear cell renal cell carcinoma and nontumorous tissues showed increased tumor signals for high-mannose-binding lectins and decreased signals for fucose- and sialic-acid-binding lectins, consistent with incomplete N-glycan maturation (54). Differences among studies are not necessarily contradictory because they examined different biological materials and analytical levels and generally focused on abundance, whereas the present analysis focuses on the distribution and Shannon diversity of glycan categories at individual proteins and glycosites.

By focusing on changes in diversity at specific glycosites, additional insights can be made into the factors involved in glycosylation shifts in renal tumor development and growth. For example, glycosites with altered diversity are enriched for complex interactor domains, consistent with the possibility that glycosylation changes influence protein-protein interactions at these sites. High-diversity regions in fibronectin are found in collagen-binding sites and, importantly, in a site important for protein folding, raising the possibility that glycosylation may alter or disrupt protein structure, as might be required to facilitate tumor cell invasion. Increasing protein folding stability and/or the variety of flexible interacting domains may reflect the physico-chemical properties of

the sugar structures attached to the proteins, which differ significantly between tumor and normal tissue. Changes in total-protein abundance were moderately associated with diversity changes but did not account for most of the linear variation, and some proteins showed higher diversity despite lower abundance. Likewise, the observation that diversity can vary across entire proteins (e.g. IGHA2, AOC3, COL6A1, POSTN, MMRN2, FN1, A2M, PLXNB2, MAN2B1, and TMED9) and at only single glycosites in other proteins (e.g. AZGP1, COL12A1, F2, HSPG2, HYOU1, ITGA1, ITGA3, ITGAV, LAMA5, ORM1, SERPINA3, SERPING1, and VCAN) indicates that diversity is site-specific and multifactorial rather than a simple consequence of increased protein expression.

Joint analysis of glycosylation-related transcript levels and diversity at specific glycosites nominated candidate associations with tumor-associated glycosylation diversity. Using uniLASSO to identify parsimonious associations, we identified MGAT1, DPAGT1, B3GNT8, ST3GAL3, and ST3GAL5 as frequently selected candidate predictors associated with discrete glycosite-glycan features. Because the glycoproteomic data were collapsed into broad glycan categories that cannot resolve linkage, core versus outer-arm fucosylation, or glycan isomers, the models cannot assign a specific enzyme to a specific structure. Increased transcript expression also does not necessarily imply increased glycosylation diversity. The Figure 7 network should therefore be read as a hypothesis-generating set of model-selected associations rather than as a causal biosynthetic map.

Our observation of increased or, less commonly, decreased glycosylation diversity in tumors that are limited to specific glycosites raises several questions. First, it is unclear why modulation of glycosynthetic enzymes should result in glycan diversity changes for only a small set of proteins and only in discrete glycosites in specific proteins. It is possible that there are structural reasons – that glycosylation alterations occur only at the sites that are exposed based on protein misfolding in tumors. It is also possible that the site-specific diversity is due to enzyme

specificity for particular glycosites. Glycan diversity at individual glycosites could lead to targeted degradation of those proteins, leading to the shift in glycoprotein levels and patterns in tumors compared to normal tissues. However, the current data do not establish which mechanism operates at any individual site. The site-resolved results instead nominate proteins and glycosites for future experiments.

A larger question is: why do many sites show increased diversity while others show decreased diversity? For the sites showing lower diversity in the tumors, it is possible that there is selective pressure favoring specific functional glycoforms. However, in tumors, most sites display greater diversity, raising the question of why diversity is selected for in tumors. Tumors could derive advantage if glycosylation changes in functional domains lead to broader activity, such as in receptor/ligand interactions. It is possible that diversity disrupts ECM structural proteins to facilitate tumor cell invasion. Diversity might also titrate the degree or affinity of receptor, ligand interactions, effectively tuning interactions in ways that are advantageous in tumors compared to normal tissues. Irrespective of the possible mechanisms, diversity can be used to nominate proteins and specific glycosites for experiments designed to determine whether and how these post-translational modifications affect tumor biology.

Current glycoproteomic technologies are more mature for detecting and quantifying heterogeneity than for establishing its functional consequences. Accordingly, the present study is partly descriptive and hypothesis-generating. Its contribution is to identify reproducible tumor-associated changes concentrated at defined proteins, glycosites, domains, and pathways and thereby narrow the candidates for mechanistic investigation.

Several limitations should be considered. The analysis is based on a single cohort and did not include independent biological workups of the glycoproteomics samples. Randomization,

blank injections, instrument-quality-control runs, and consistent processing reduce the likelihood that the observed patterns are analytical artifacts but do not replace independent replication. The Shannon index is internally scaled to relative glycan-category proportions rather than absolute glycopeptide spectrum-match counts, although data-dependent acquisition, sampling depth, and protein abundance can still affect detection of low-abundance glycoforms. Broad glycan categories provide limited structural resolution, and RNA-seq associations do not establish enzymatic causality. Notably, in a previous prostate cancer glycoproteomics study (30), a more descriptive analysis did yield similar observations of increased glycan heterogeneity on specific proteins and glycosites in tumor compared to adjacent non-cancer tissue. Future studies should include targeted glycopeptide quantification, independent cohorts and sample reprocessing, perturbation of candidate glycosylation enzymes, site-directed mutagenesis, and protein-specific functional assays.

In summary, deep glycoproteomic analysis of renal tumors has revealed glycan diversity at specific glycosites as the most striking difference between renal tumor and normal tissues. This diversity can be captured using Shannon's diversity index, a useful metric for identifying key regions of glycosylation changes. The largest changes were highly site-specific, cluster in functional and structural protein domains, and focus on specific functional pathways, most prominently extracellular and immune proteins. Integrative RNA-seq modeling identified candidate transcript-glycosite associations for future testing. Independent validation and mechanistic experiments will be required to determine the functional consequences and potential biomarker value of the nominated glycosylation changes.

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DATA AVAILABILITY

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE (41) partner repository with dataset identifier PXD075353. The submission includes raw mass spectrometry files, Byonic result spreadsheets, search parameters, the focused protein FASTA database, the N-glycan library, sample-to-file mappings, and available native Byonic result (.byrslt) and spectra (.byspec2) files; the exceptions and verification status are described in Methods. Supplementary Table S2 provides the filtered glycopeptide-spectrum matches and identification metrics, Supplementary Table S3 provides glycosite-level Shannon diversity and Benjamini-Hochberg-adjusted statistics, Supplementary Table S4 provides total-protein abundance measurements, and Supplementary Table S5 provides the glycosylation-related RNA-seq analysis. Gene-expression data have been deposited in Gene Expression Omnibus (GEO accession no. GSE325575).

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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FIGURES

Figure 1. Dataset Overview. (A) The percentage of glycopeptide spectral matches mapped to each of the seven glycan categories in both renal tumors and normal-adjacent tissue. For each glycan category, an example glycan structure is depicted. Error bars represent the standard deviation of the glycan categories measured across the 32 patients. Blue squares represent N-acetylglucosamine, green circles represent mannose, yellow circles represent galactose, the purple diamond represents N-acetylneuraminic acid, and the red triangle represent fucose. Abbreviations: GSMs = glycopeptide spectral matches; Fuc = complex fucosylated; Fuc + Sia = complex fucosylated and sialylated; Sia = complex sialylated; Undec = complex undecorated; Mann = high mannose; IC = incomplete core. **(B)** Box plot of glycosylation diversity at the whole glycoproteome, compared to **(C)** box plot of glycosylation diversity at the individual glycoprotein level, each in renal tumors and normal-adjacent tissue. The horizontal lines in the boxes represent the median Shannon diversity indexes, and each dot represents one glycoprotein. The means were compared using a Mann-Whitney U-test; ***** $p < 0.00001$, n.s. $p \geq 0.01$. Blue data represents renal tumors, and orange data represents normal-adjacent tissue.

Figure 2. Exploratory Protein-Level Glycosylation Diversity Analysis. (A) Heatmap of 100 proteins meeting a raw Mann-Whitney U p-value < 0.01 between renal tumors and adjacent normal tissue. Eighty-six proteins had higher diversity and 14 had lower diversity in tumors. Raw p-values were used only to rank features for this exploratory display; Benjamini-Hochberg-adjusted p-values were used for multiple-testing inference in the text. Rows represent proteins, and columns represent glycan categories. Within the heatmap, darker red indicates more glycopeptide spectral matches (GSMs) per protein in that glycan category. For the change in diversity, dark red

indicates a greater increase in diversity in renal tumor tissue, and darker blue represents a greater decrease in diversity in renal tumor tissue. **(B)** Overrepresentation of Gene Ontology Biological Process terms from the 86 proteins with increased glycosylation diversity in renal tumors compared to normal-adjacent tissue in **(A).** The fold enrichment of each term is plotted as a function of its significance, and the data points are sized according to the number of proteins in the dataset that match to that term. Terms related to extracellular structure organization, the immune response, and cell adhesion are highlighted in red, and the most significant terms in these three categories are labeled. (**C**) Based on Pearson correlation, the glycan category distributions for these 100 glycoproteins in (**A**) were clustered. Clusters with the largest number of glycoproteins (clusters #3 [light green], #5 [light purple], and #6 [light blue]) are shown in more detail. Abbreviations: GSMs = glycopeptide spectral matches; Fuc = complex fucosylated; Fuc + Sia = complex fucosylated and sialylated; Sia = complex sialylated; Undec = complex undecorated; Mann = high mannose; IC = incomplete core.

Figure 3. Glycosylation Heterogeneity Differences by Tissue Type. (A) Heatmap of protein-level Shannon diversity for features quantified in >= 40% of samples, with renal tumors on the left and adjacent normal tissues on the right. Rows refer to glycoproteins, and columns refer to samples. Dark red indicates an increase in glycosylation diversity. (B) Volcano plot of tumor-versus-adjacent-normal differences in protein-level Shannon diversity across 177 proteins. Volcano axes show raw p-values for visualization, whereas Benjamini-Hochberg-adjusted p-values were used to control the false discovery rate in inferential analyses. Red and blue indicate higher and lower diversity in tumors, respectively, and gray indicates features not meeting the adjusted significance threshold. Selected proteins are labeled; yellow-highlighted labels indicate

an association between diversity and renal tumor stage. (C) Volcano plot of tumor-versus-adjacent-normal differences in glycosite-level Shannon diversity, highlighting selected sites on FN1, MAN2B1, and MMRN2. The same raw-p-value visualization and adjusted-inference convention was used as in panel B. Selected glycosites are labeled as gene name and N-glycosite.

Figure 4. Glycosylation of Fibronectin (FN1). (A) Graphical representation of the amino acid sequence of FN1 and its domains and possible N-glycosites. Blue ovals represent the FN type-I domain, red ovals represent the FN type-II domain, green diamonds represent the FN type-III domain, the purple hexagon represents the FN type-III 8 domain/extra domain B, and the yellow tab represent the FN type-III 13 domain/extra domain A. Circles represent N-linked glycosites annotated in UniProt. Labeled glycosites were identified with ≥2 GSMs. **(B)** Heatmap of the eight glycosites with ≥ 2 glycan compositions. Rows refer to glycan compositions, and columns refer to samples (renal tumors on the left and adjacent-normal tissue on the right). On the left side of the heatmap, the glycan type is represented with one of seven colors: purple indicates complex sialylated, red indicates complex fucosylated, blue indicates complex fucosylated and sialylated, green indicates high mannose, hybrid indicates orange, yellow indicates complex undecorated, and gray indicates incomplete core glycans. Putative structures of example glycan compositions in renal tumors are highlighted on the left of the heatmap.

Figure 5. Glycosylation of UDP-glucoronosyltransferase 1A9 (UGT1A9). (A) Donut plot comparison of glycopeptide spectral matches (GSMs) for UGT1A9 in renal tumors (left) and normal-adjacent tissue (right). **** indicates $p < 0.0001$. Abbreviations: GSMs = glycopeptide spectral matches; Fuc = complex fucosylated (red); Fuc + Sia = complex fucosylated and sialylated

(blue); Sia = complex sialylated (purple); Undec = complex undecorated (yellow); Mann = high mannose (green); IC = incomplete core (gray); Hybrid = hybrid (orange). **(B)** Heatmap of two glycosites on UGT1A9 that were identified in both renal tumors and normal-adjacent tissue: N71 and N344. Rows refer to patients and show data from N71 at the top and N344 at the bottom. Rows are shown in increasing order of cancer stage. Columns refer to glycan category, with renal tumors on the left and adjacent-normal tissue on the right. Darker red indicates higher GSMs at that glycosite. **(C)** Putative representations of glycan compositions identified with ≥ 2 GSMs are illustrated in overlapping Venn diagrams to illustrate which glycans were identified in renal tumors and those identified in normal-adjacent tissue for N71 (top) and N344 (bottom). Blue boxes represent renal tumor glycans, and orange boxes represent normal-adjacent glycans. Blue squares represent N-acetylglucosamine, green circles represent mannose, yellow circles represent galactose, purple diamonds represent N-acetylneuraminic acid, and red triangles represent fucose.

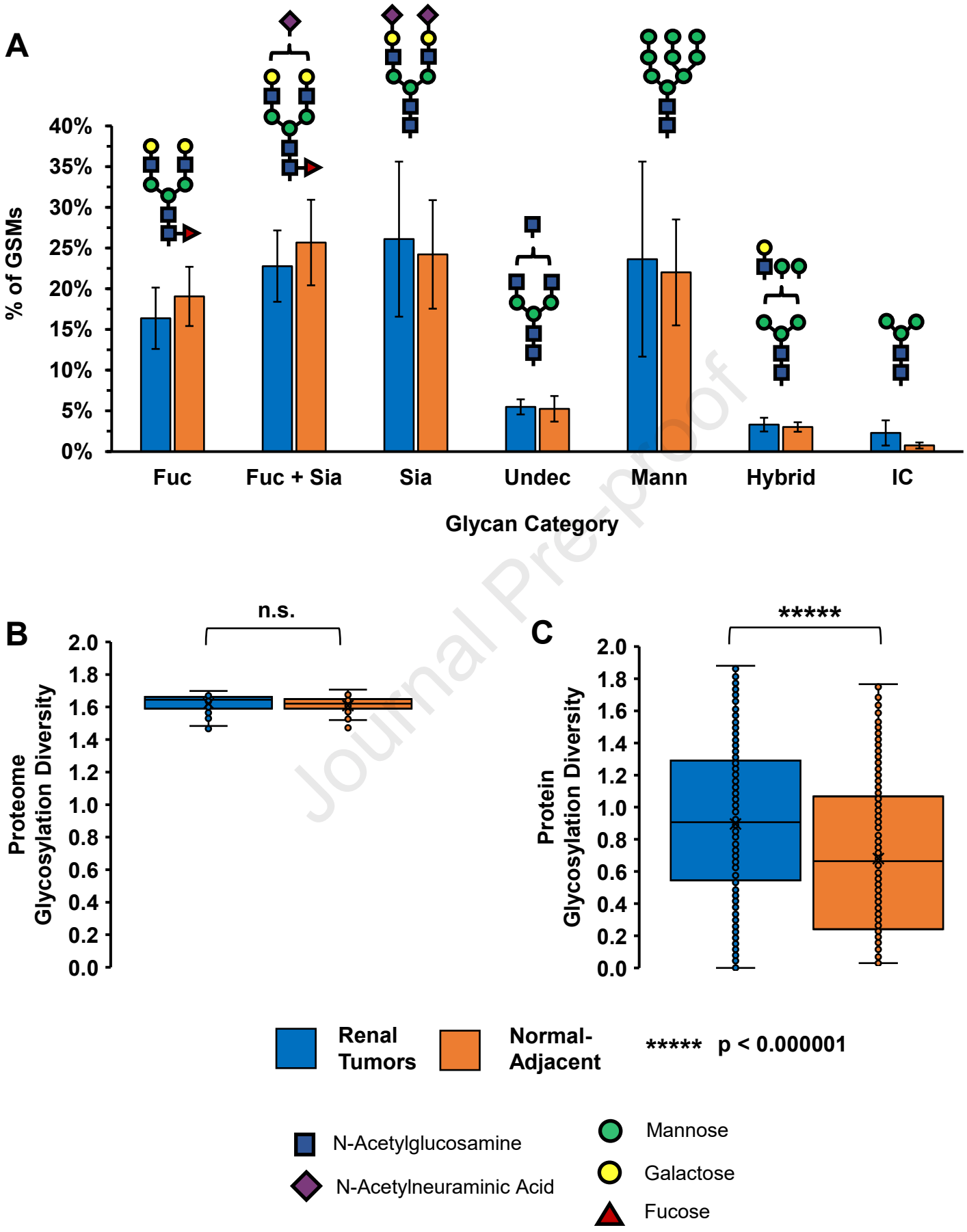
Figure 6. Levels of Glycosylation-Related Transcripts by RNA-seq. (A) Box plots of log10 transcripts per million (TPM) for FUT6, FUT8, and GFUS in renal tumors (blue) and adjacent normal tissue (orange). GFUS encodes GDP-L-fucose synthase and is included as a fucosylation-related transcript because it contributes to GDP-fucose biosynthesis, although it is not a fucosyltransferase and was not included in the downstream RNA-seq integration analyses. FUT11/POFUT4 was excluded because it is a protein O-fucosyltransferase and cannot be assigned to the N-glycan categories analyzed here. (B) Hypothetical complex fucosylated and sialylated glycan showing plausible enzymes for selected linkages; this schematic does not assign causal enzymes to the measured glycan categories. (C) Box plots of TPM z-scores for FUT8, DPAGT1, ALG1, MGAT1, and ST3GAL2. ST3GAL1 showed only a small, non-significant change and is

not displayed as the significant ST3GAL example. Horizontal lines indicate medians and points indicate samples. Asterisks indicate Benjamini-Hochberg-adjusted significance: * adjusted $p < 0.05$, ** adjusted $p < 0.01$, *** adjusted $p < 0.001$, **** adjusted $p < 0.0001$, and ***** adjusted $p < 0.00001$.

Figure 7. Sankey Diagram of Enzyme-Transcript-Glycan-Type-Glycosite Associations. The diagram summarizes model-selected associations between glycosylation-related transcripts from RNA-seq and glycan-type-glycosite features from glycoproteomics. Glycan-type-glycosite pairs were displayed when selected in > 3% of uniLASSO iterations, and enzyme transcripts were displayed when selected in > 1% of iterations. Connection widths indicate feature-selection frequency. The network is intended to by hypothesis-generating and prioritize candidate associations for future testing and does not represent a definitive biosynthetic pathway or establish direct enzymatic causality.

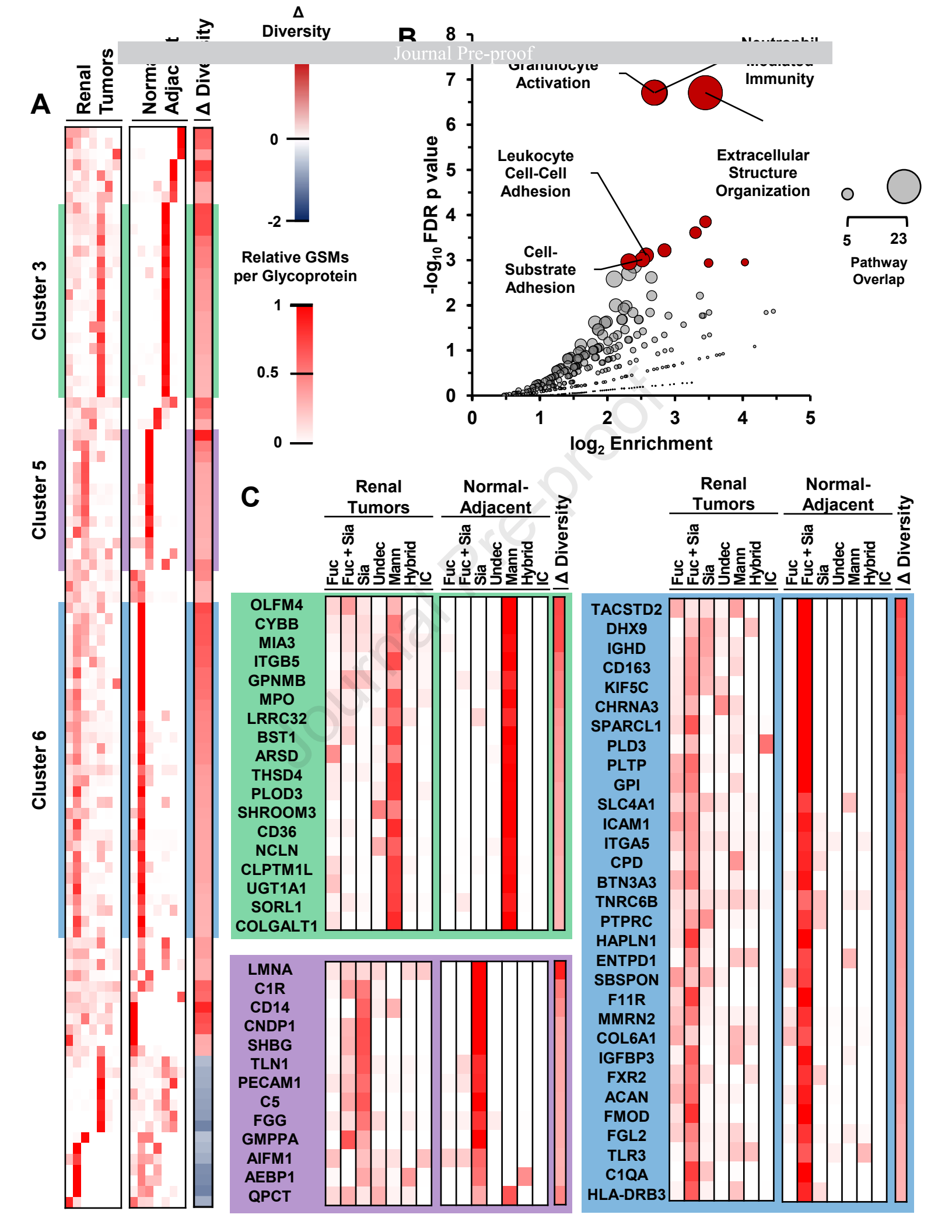
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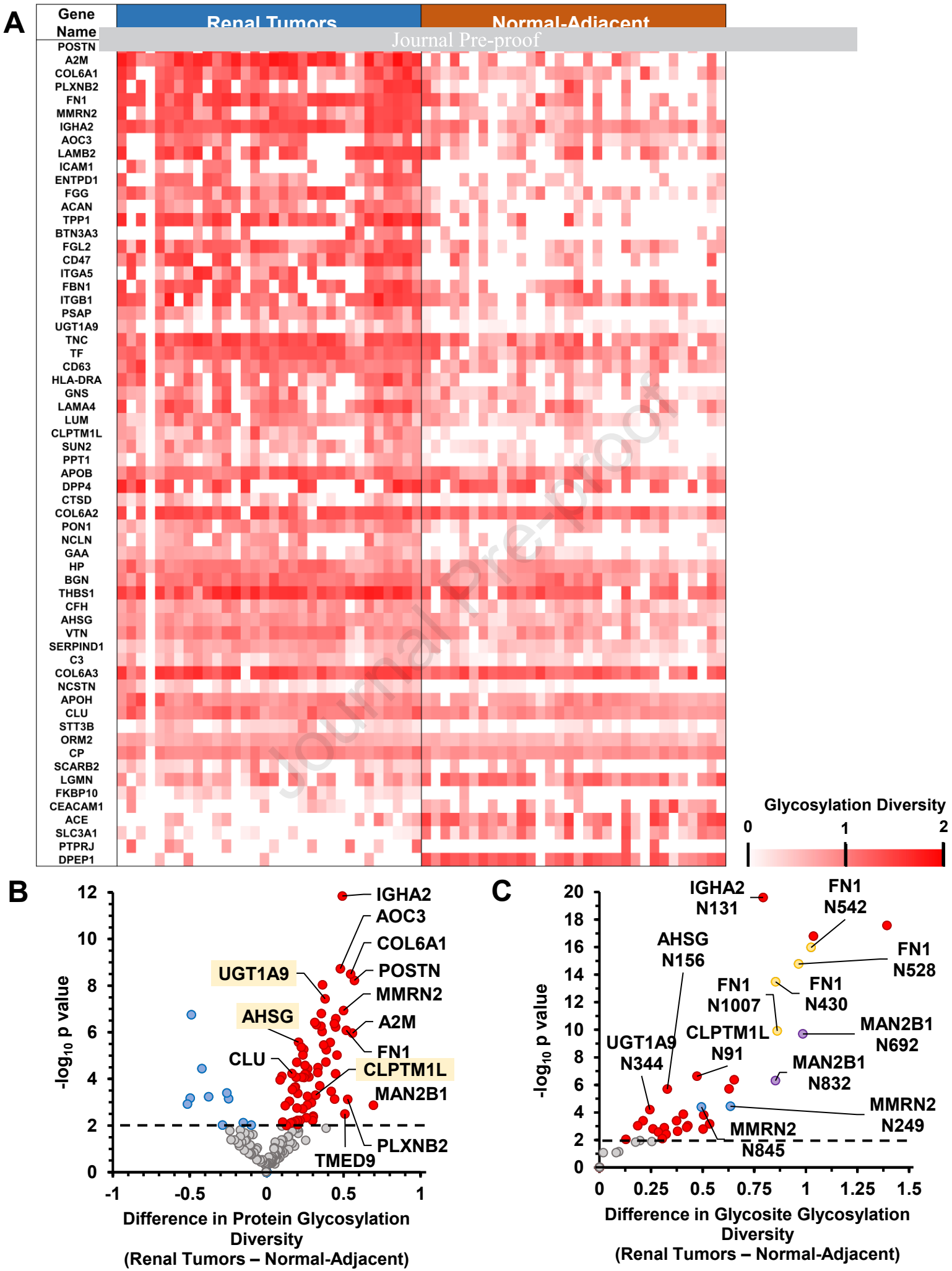


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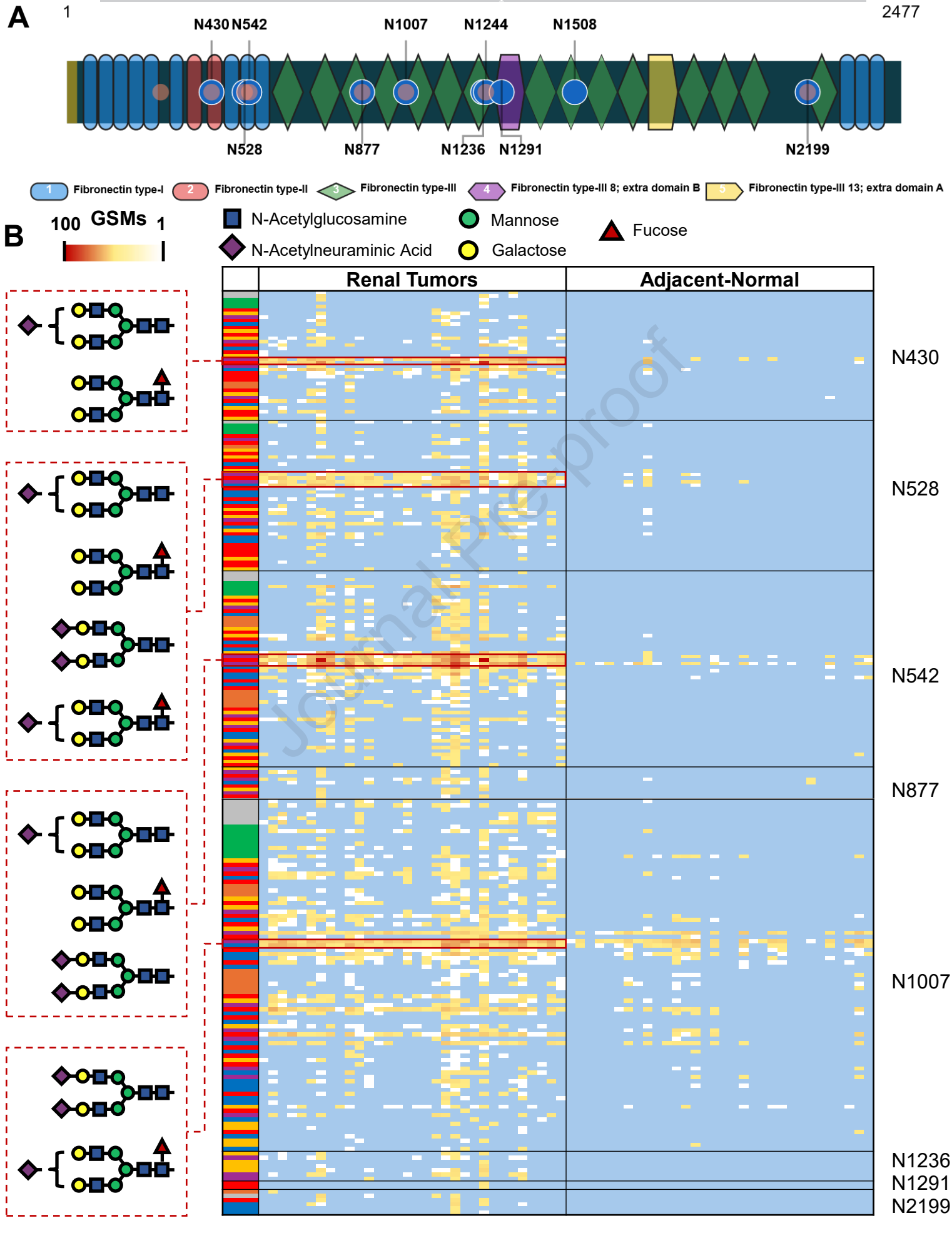
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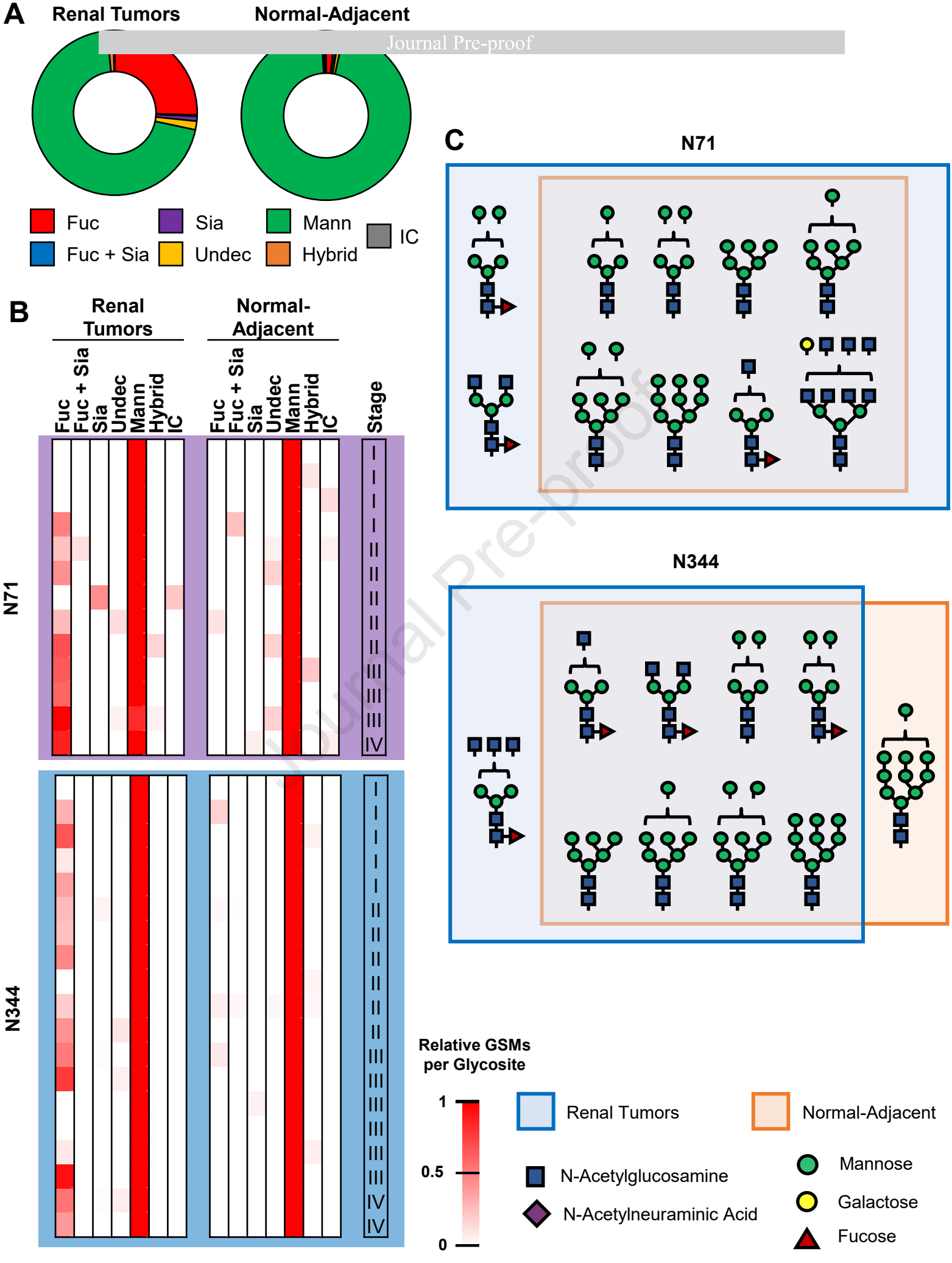


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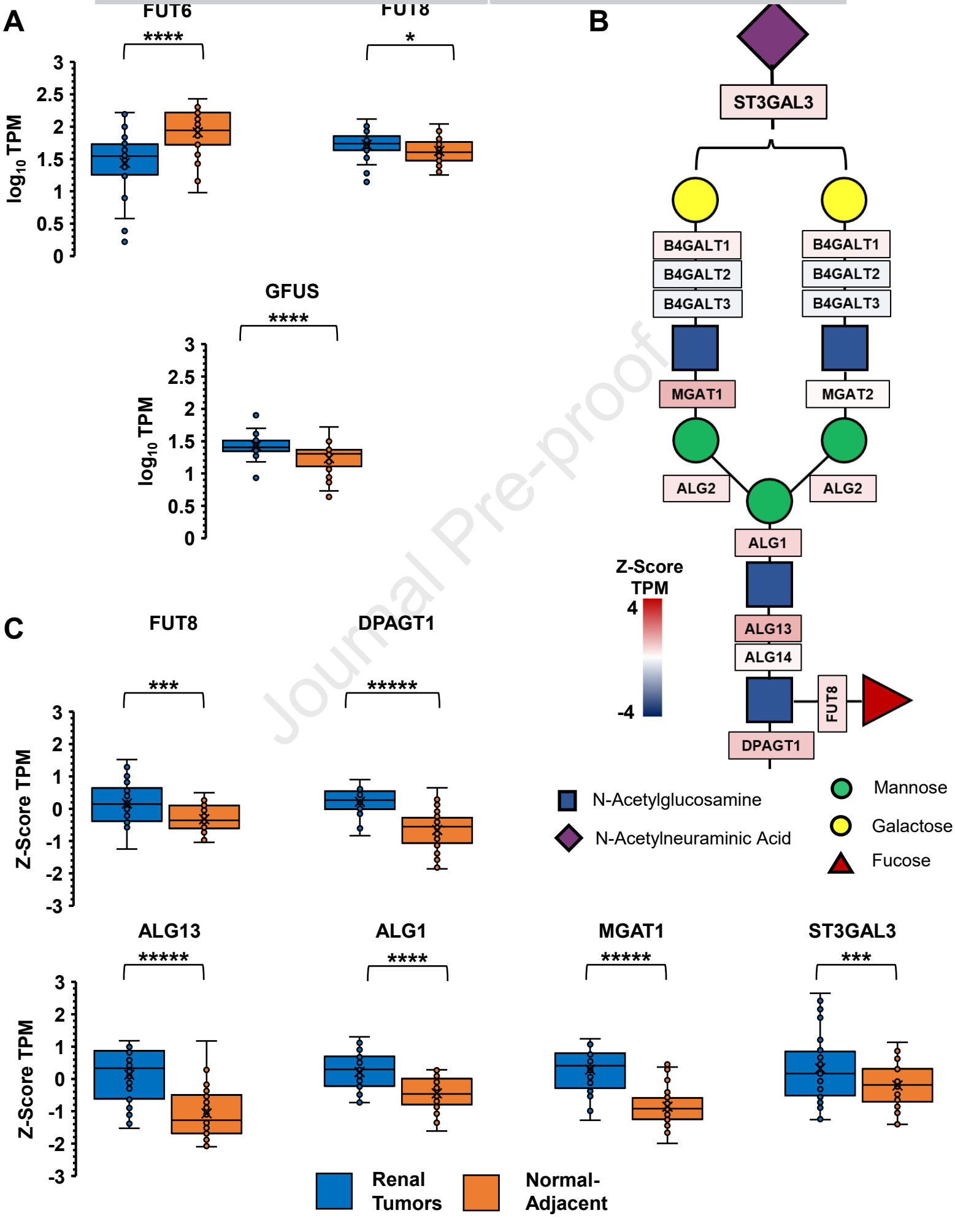


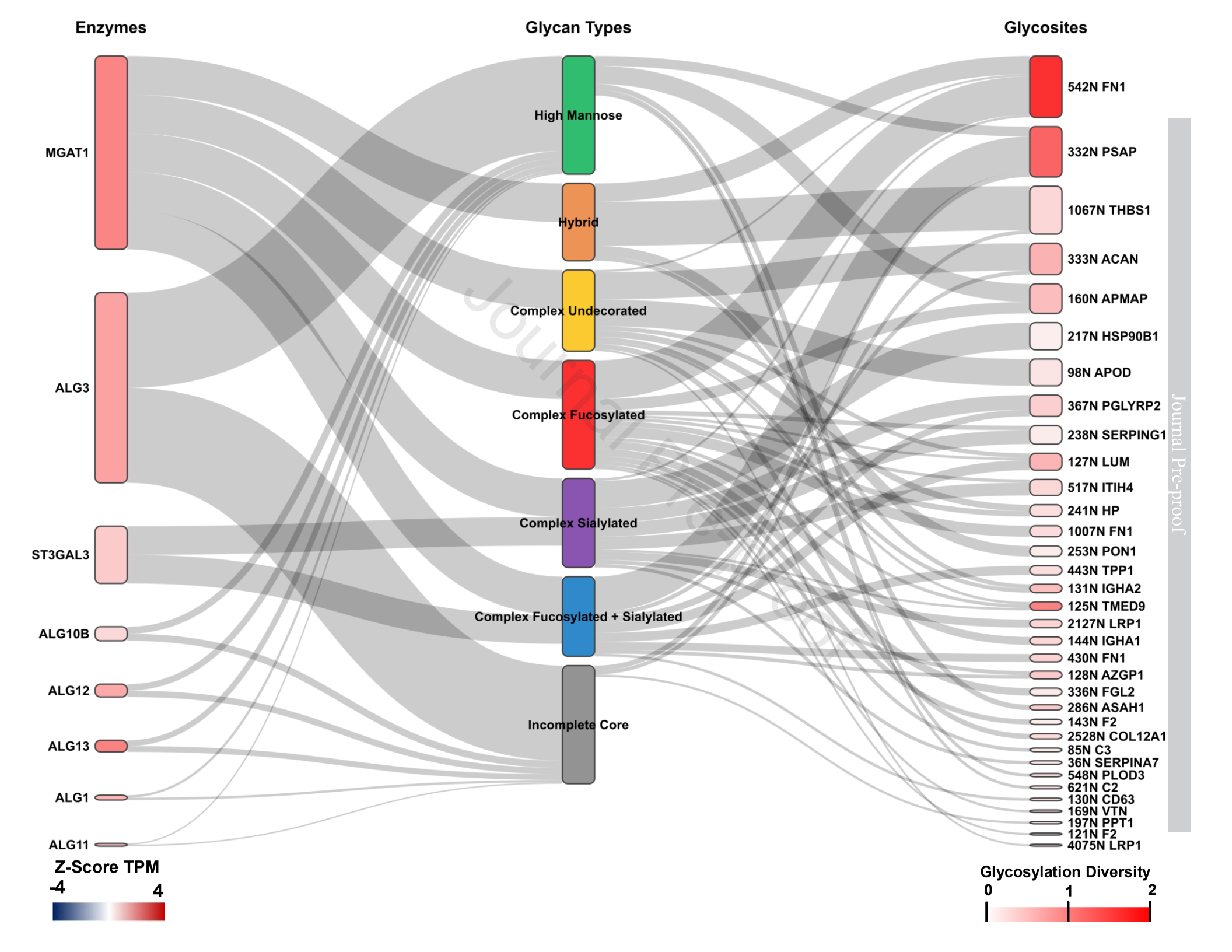
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Enzymes
Glycan Types
Glycosites
MGAT1
ALG3
ST3GAL3
ALG10B
ALG12
ALG13
ALG1
ALG11
High Mannose
Hybrid
Complex Undecorated
Complex Fucosylated
Complex Sialylated
Complex Fucosylated + Sialylated
Incomplete Core
542N FN1
332N PSAP
1067N THBS1
333N ACAN
160N APMAP
217N HSP90B1
98N APOD
367N PGLYRP2
238N SERPING1
127N LUM
517N ITIH4
241N HP
1007N FN1
253N PON1
443N TPP1
131N IGHA2
125N TMED9
2127N LRP1
144N IGHA1
430N FN1
128N AZGP1
336N FGL2
286N ASAH1
143N F2
2528N COL12A1
85N C3
36N SERPINA7
548N PLOD3
621N C2
130N CD63
169N VTN
197N PPT1
121N F2
4075N LRP1
Z-Score TPM
-4
4
Glycosylation Diversity
0
1
2
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HIGHLIGHTS

- Intact glycoproteomics analysis of patient renal tumors and normal tissue.
- Shannon diversity index applied to quantify glycosylation heterogeneity.
- Multilevel findings on site-specific glycosylation and heterogeneity.
- Glycosylation changes in biological pathways and protein domains.
- Suggested links between glycosylation transcripts and site-specific diversity.

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HIGHLIGHTS

Glycosylation, the attachment of sugars to proteins, changes in disease. We developed an approach using the Shannon diversity index to quantify glycosylation diversity across proteins and individual glycosites. Analysis of 32 paired renal tumors and adjacent normal tissues revealed localized diversity differences associated with specific protein domains and proteins involved in cell adhesion and immunity, which protein abundance only partly explained. Integration with gene expression identified candidate associations, providing a framework for prioritizing glycosylation changes for future tumor biology studies.

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Declaration of Interest Statement

☑ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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